Mapping the Shape of the U.S. Economy: A Topological Data Analysis Approach with Ball Mapper

Ryan Johnson[*]

Graduate, Department of Mathematics, University of Alaska Anchorage, Anchorage, AK

Student: *johnson.ryan-0@pm.me*[*]
Mentor: *scook25@alaska.edu*

KEYWORDS

Topological Data Analysis; Topology; TDA; Ball Mapper; Mapper; Data Science; Economics; Macroeconomics;

ABSTRACT

Topological Data Analysis (TDA) is a developing data analysis method. Currently, a large body of TDA research utilizes the traditional Mapper algorithm. This paper aims to show an entry level path into TDA, and to expand the body of literature using Ball Mapper–an algorithm adjacent to traditional Mapper. Using U.S. macroeconomic data from the Bureau of Economic Analysis (BEA), the Federal Reserve Bank (FRB), and the Bureau of Labor Statistics (BLS), we begin by showcasing Ball Mapper's use in exploratory data analysis. We then show how we can effectively use Ball Mapper in combination with other derived economic measures: the Real Interest Rate, PPI-CPI Gap, and Wage Growth. Our results reveal an underlying structure in the U.S. economy from 1960 to 2024. Illustrated through Ball Mapper's graphs, we find that economic expansion holds distinct stages, that peaks and troughs are structurally separate from the broader expansion cycle, and that no two economic contractions are alike. Notably, despite not being included as an input, our analyses uncovered meaningful relationships between years. Since Ball Mapper revealed time-related relationships without relying on temporal data, this suggests that the relationships seen are inherent to the economic data itself.

INTRODUCTION

Topological Data Analysis (TDA) is an emerging field of data analysis that offers a unique vantage point for exploring complex data. Broadly, traditional TDA applications use two tools: an algorithm called Mapper whose output is represented by a mathematical graph and an analysis technique called Persistent Homology.[1][A] These two tools are not necessarily dependent on one another and both provide information about data in their own right. Moreover, Mapper and Persistent Homology are central to TDA's core argument: data contains an underlying shape, and this shape can provide us with qualitative, and sometimes quantitative, insights about large multidimensional datasets.[2]

At a high level, Mapper is used to explore and visualize high-dimensional data. Briefly, the Mapper algorithm achieves this by producing a mathematical graph using a filter function. This graph presents a picture analogous to taking an x-ray of the data, revealing the skeleton of the dataset; it allows one to find structural relationships that are not obvious at first glance. Persistent Homology's goal is to uncover topological features of a dataset (components, holes, voids, etc.) through various scales. A helpful way to think about this is similar to viewing a small picture made of dots as you move it from touching your nose to arm's length away. Too close, you only see individual dots; too far away, the dots merge together into a singular shape. Persistent Homology takes observational notes about how the picture changes across all the various distances.

Moving forward, we will use the terms "traditional Mapper" and "Mapper" interchangeably. When we use the term, "traditional Mapper" we are referring to methodology for Mapper presented by Singh, et al.[3] in 2007. This is an important distinction because the concept of the Mapper methodology has been implemented in various ways by changing the filter function and other parameters.

Although Persistent Homology provides insight into the topological features our data contains, the focus of this paper is on the structural exploration that Mapper and Mapper variations offer. Specifically, we are examining various macroeconomic indicators of the

[A] A mathematical graph, from Graph Theory, is a diagram that consists of nodes and edges.

United States (U.S.) using a Mapper-adjacent algorithm called Ball Mapper (BM). Ball Mapper is of particular interest to us because it requires fewer parameters than traditional Mapper, allowing for easier implementation.

In traditional Mapper, one must pass the data through three stages, all with their own parameters, in order to generate a graph. Ball Mapper, however, only needs one parameter before producing a graph.[4] An important trade-off between Ball Mapper and traditional Mapper is that while BM reduces the number of steps, one loses control of being able to fine-tune outputs. Additionally, traditional Mapper literature presents itself as requiring a mathematical background, such as an introductory course in Topology, or conceptual knowledge of data science methods and algorithms. Although it is not impossible to learn TDA without the above backgrounds, TDA might seem unnecessarily complicated for data analysis, or impossible to decipher.

One major motivation for this paper came from a paper by Dłotko, et al.[5] The paper was a two-pronged macroeconomic analysis using Ball Mapper. The first prong used a dataset of five macroeconomic variables from 16 countries from the 1870s to 2017. The authors were interested in comparing how countries have evolved over time, their transformation from the Great Depression Era, and various views on wealth and inequality. The second prong was to look at the relationship between private credit growth and the GDP of various countries. This paper was the only one in the literature that we could find which combines both macroeconomics and Ball Mapper.

If successful, this analysis will expand the small body of literature whose main focus is applications with Ball Mapper.[1] We also hope that it will serve as an on-ramp for anyone interested in TDA who feels inundated with jargon during early-stage research.

Using U.S. macroeconomic data from the Bureau of Economic Analysis (BEA), the Federal Reserve Bank (FRB), and the Bureau of Labor Statistics (BLS), we plan to achieve these objectives in two stages. The first stage is to show Ball Mapper's use in exploratory data analysis (EDA) by looking at structural observations, coloration, as well as the size variation of nodes in our graph. Then, using what we demonstrated in our first stage, we extend that framework to other derived macroeconomic measures: the Real Interest Rate[B][6,7] PPI-CPI Gap,[C][8] and Real-Wage Growth.[9][D]

METHODS AND PROCEDURES

Data Selection & Preparation

We pulled data from three publicly available data from U.S. government sources: The Bureau of Economic Analysis (BEA), The Federal Reserve Board (FRB), and The Bureau of Labor Statistics (BLS).[10,11] The data were gathered using R using two application programming interfaces (APIs): one for data from the BEA, and the other from the Federal Reserve Economic Data (FRED) API.[12,13][E] We additionally used recession dates based on business cycle contractions and expansions provided by the National Bureau of Economic Research (NBER). These agencies were selected because of they are authoritative sources for U.S. economic data. Their widespread use in both the private and public sectors gives us high confidence in the accuracy and integrity of the data.[14]

Data Series	Series Abbreviation	Year Start	Year End	Frequency	Series	API
Gross Domestic Product	GDP	1930	2024	Annual	Table 1.1.1	BEA
Personal Income & Its Disposition	PID	1948	2024	Annual	Table 2.1	BEA
Foreign Direct Investment	FDI	2014	2024	Annual	Direct Investment and MNE	BEA
Federal Funds Rate	FFR	1955	2025	Annual	RIFSPFFNA	FRED
Employment Cost Index	ECI	2001	2025	Quarterly	ECIALLCIV	FRED
Consumer Price Index	CPI	1947	2025	Monthly	CPIAUCSL	FRED
New Privately-Owned Housing Units Started	Housing Starts	1959	2025	Monthly	HOUST	FRED
Producer Price Index - Finished Goods	PPI	1947	2025	Monthly	WPSFD49207	FRED
Unemployment Rate	Unrate	1948	2025	Monthly	UNRATE	FRED

Table 1: Summary of data series, time period, and sources for this analysis.

Table 1 shows all the data series we considered for this analysis. We excluded Employment Cost Index (ECI) and Foreign Direct Investment (FDI) due to their limited availability of years. Of the remaining data series, New Privately-Owned Housing Units Started

[B] Approximation using the Fisher Equation.
[C] Producer Price Index: Finished Goods less Consumer Price Index. Details on data transformation in the Methods and Procedures section.
[D] Compensation less Inflation (CPI). Details on data transformation in the Methods and Procedures section.
[E] FRED aggregates data from national and international sources, as well as public and private sources.

(Housing Starts) had the smallest range and so set our lower bound for the range years used (1960-2024). Our upper bound was the latest full calendar year that was available. We wanted to represent the economy from various vantage points. i.e. the broad economy, policy makers, consumers, business owners, etc. Table 2 shows the data series that we felt were representative of the economy as a whole and had data for our time period of interest. Within each of these series we used a subset of columns to construct our final dataset for analysis. To see what parts of the economy we determined each series plays, the column Functional Description, can be found Table 2.

For this analysis we are focusing on an annual time frame and some data transformation was needed. Looking at Table 1, you can see that not all our data was provided on an annual basis. Table 2 shows a summary of the transformation taken and the details for each follow.

The first transformation we did was on Personal Income and Its Disposition (subsequently, Personal Income) is reported in nominal dollars. Since nominal dollars do not account for inflation, we used the Consumer Price Index (CPI) we transformed our values to Real Dollars (Equation 1).[15] We then took the log difference (Equation 2) from the output from Equation 1 to get an approximate percentage change from the preceding year.

$$\text{Real Dollars} = \frac{\text{Nominal Dollars}}{\text{CPI}} \times 100 \tag{1}$$

Calculating the year-over-year percent change using log differences provides us with a few advantages. First, taking the log stabilizes the variance and makes exponential trends across the series more linear. Second, the difference between log values approximates the true percentage change well for small to moderate values. Further, the log difference allows for symmetric percent changes: increases or decreases from year to year are of equal magnitude.[16] An interesting question that has not been explored to our knowledge is the impact on linearized data and Ball Mapper's distance metric, the Euclidean distance. Further exploration of this question will be touched on in the discussion section.

$$\Delta \ln(\text{Level}) = [\ln(\text{Level}_t) - \ln(\text{Level}_{t-1})] \times 100 \tag{2}$$

Our next transformations were applied to housing starts, Producer Price Index - Finished Goods (PPI), and CPI. All three of these sources were only provided on a monthly time frame, so to get an annual value for them we used the simple arithmetic mean. Once these data were in annual form, we then found the percent change from the previous year using the log difference (Equation 2).

Housing starts data is given in a seasonally adjusted annual rate (SAAR). To get this number the U.S. Census first takes the raw monthly number of housing starts reported to their survey. Then they adjust that number to account for seasonal patterns (e.g. summertime booms or winter lulls), and finally they multiply it by 12. So how we interpret this final number: for the reporting month, it is the number of houses that would be built at the end of a 12-month period.[17]

Series Abbreviation	Economic Role	Functional Description	Transformation Applied
GDP	Growth Composition	Consumption, Investment, Gov., and Trade	None (Source in % Change)
PID	Income Structure	Wages, Entrepreneurship, Gov. Transfers	Real Adjustment & Log Difference
UnRate	Labor Market Dynamics	Labor Distress Level and Trend	None (Annual Rate) & Simple Difference
FFR	Monetary Policy	Cost of Capital and Borrowing Conditions	None (Annual Rate) & Simple Difference
PPI	Supply-Side Signal	Producer Costs & Measures Cost-Push Inflation	12-month Avg. & Log Difference
CPI	Demand-Side Signal	Cost of Living & Measures Demand-Pull Inflation	12-month Avg. & Log Difference
Housing Starts	Leading Indicator	Physical Residential Production	12-month Avg. & Log Difference

Table 2: All years for analysis are 1960-2024.

Both PPI and CPI are economic indexes and represent a weight average of the certain good they respectively track. Although PPI has many different series, the one use for this analysis tracks the price the producers receives from the physical goods that it sells to business or consumers. An important distinction to note about the PPI used here, is that the index captures goods which are finished or final in the economic sense. Colloquially, it is similar to the wholesale price a retailer might pay to a producer before marking up for retail sales. However, it also includes goods which are sold to business, think fully assembled computers or fleet vehicles.[18] CPI on the other hand captures the consumer side and the retail price. That is, the cost of the good after profit margin, transportation cost, and other intermediary costs are calculated into a goods price sold at the register.[19] Broadly, both have been associated with measuring inflation for

producers and consumers. The more nuanced understanding is that index changes can reflect many things: push-cost inflation, demand-pull inflation, supply shocks, monetary policy inflation, and wage-price spirals are just a few examples. It is our hope that Ball Mapper might show the distinctive separation in the various aforementioned reasons for the PPI and CPI indexes changing from year to year. [SAM: I am worried about this ending sentence and that if I say it here I will need to do it for all the other data.]

Finally, after all the transformations are complete, we used a total of 16 variables from our selected data series (Table 2) for our analysis. Again, we aimed to use a comprehensive set of variables to represent the U.S. economy. A list of variables use can be found in Table 3 in no particular order.

PPI Finished Change	Personal Income	Compensation	Entrepreneurship
Transfers	Consumption	Domestic Investment	Government Spending
Exports	Imports	Inflation	Unemployment
Unemployment Change	Fed Rate	Fed Rate Change	Housing Change

Table 3: Variables inculde in Ball Mapper Point Cloud

Ball Mapper

In depth descriptions of Ball Mapper's theory have been covered in various papers.[4,20,21] As the aim of our paper is to provided a introductory foray into Ball Mapper and TDA in general, we will omit some of the more technical details below but the previously mentioned references provide a more in-depth explanation.

Theoretically, Ball Mapper supposes we are given a dataset X with K dimensions, N observations and some metric d. Then to create a Ball Mapper graph, our goal is to construct a collection of balls, let's call it B, with a constant radius, ϵ, such that every observation of our dataset is covered by at least one ball in B. Mathematically, this is written as follows: Let (X, d) be a metric space where X is our dataset and d is a distance metric on X. For a fixed radius $\epsilon > 0$, let $C \subseteq X$ be a set of center points. Then for each $c \in C$, we define a ball $b(c, \epsilon) = \{x \in X : d(c, x) < \epsilon\}$. Our collection then is $B = \{b(c, \epsilon) : c \in C\}$ and $\bigcup_{b \in B} b = X$.

In practice, our dataset X is called a Point Cloud. Our N observations or rows are normally referenced as points, and our K dimensions, the columns of X, are denoted variables. Since Ball Mapper is know to handle large multidimensional datasets well, it is common to have point clouds with dimensions $K > 2$. [Think I need a ref. here about BM and large datasets?] Ball Mapper was originally written in R, but later on was created for use in Python. For this analysis we used R and the Ball Mapper package aptly named "BallMapper".[22]

To construct the Ball Mapper (BM) graph, the balls of our graph are created in an iterative manner and not a predetermined manner like the theory defines. If we pre-defined a subset of ball centers, it would take away from the driving idea behind TDA–that data has shape–as well as start moving into the territory of other clustering methods such as K-means where you selected a number of clusters based on statistical properties of the data. This iterative process allows our point cloud to cluster in a "natural" way based on on our choice on epsilon and distance metric.

$$d(x_i, x_j) = \sqrt{(x_{i_1} - x_{j_1})^2 + (x_{i_2} - x_{j_2})^2 + ... + (x_{i_k} - x_{j_k})^2} < \epsilon \tag{3}$$

The iterative ball creation is carried out in a linear fashion from the first row to the last of the point cloud. The result of this linear selection process is a pseudo-random creation of ball centers. How this works is that it takes the first point, let's call it c_1, of the point cloud and assigns it as the center of the first node in our graph. Then using the default distance metric, the Euclidean Distance (Equation 3), it compares every point in the point cloud to c_1 to create $b_1 = b(c_1, \epsilon) = \{x \in X : d(c_1, x) < \epsilon\}$. After this first ball is created, Ball Mapper looks for the next row in the point cloud that is not a member of the first ball, and assigns it as the next center. This process continues until all the points are in a ball and the union of balls consists of our whole point cloud.

Hopefully, it is clear now why this process can be described as pseudo-random because it relies a great deal on the point cloud as well as the epsilon. In fact, the Ball Mapper graph changes for epsilons sufficiently large enough to change at least one point's inclusion or exclusion to a ball; the graph can also change for unique orders of the point cloud rows. This is touch on in greater detail in the Discussion.

The last step to the Ball Mapper graph is to draw the edges if they exists. The way the BM algorithm is structured, it allows for the possibility of points being in multiple balls–also called nodes. This intersection of points between nodes is what creates an edge, and

the number of share points between nodes indicates an edge strength. So, once we have all the nodes, edges, and edge strengths known, we then can construct a Ball Mapper graph. The Ball Mapper algorithm described above can be broken down as follows:

1. Select a point $x_i \in X$ and assign it as c_i.
2. Given $\epsilon > 0$, construct a ball $b_i(c_i, \epsilon)$ with a center c_i and radius ϵ by checking if x_j satisfies $d(c_i, x_j) < \epsilon : x_j \in X$ and d is the distance metric (Equation 3).
3. Place b_i in a set B.
4. Select another x_n and repeat step 1-3 such that $x_n \neq x_j : x_j \in b_i$.
5. Draw an edge between $b_i, b_j \in B : b_i \neq b_j$ if $b_i \cap b_j \neq \emptyset$, weighting the edge based on the size of the intersection.

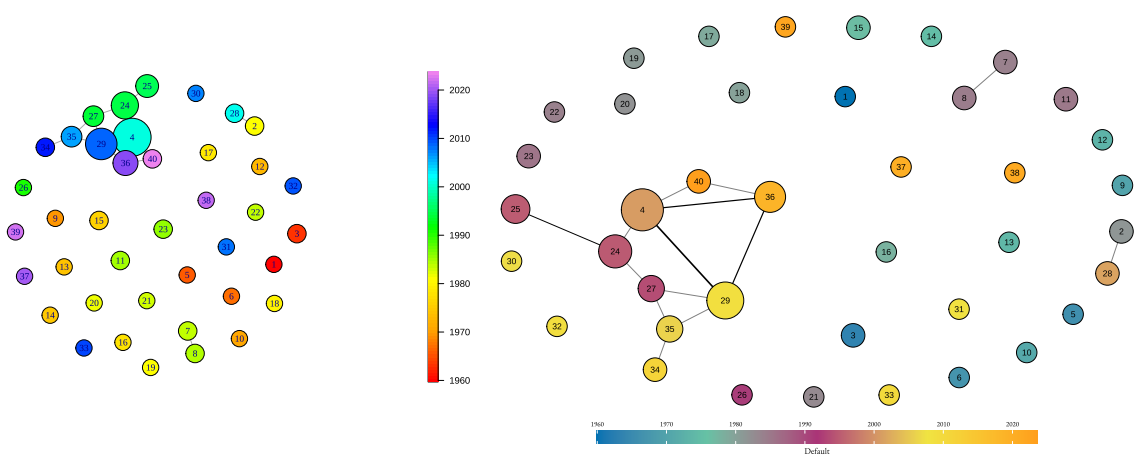
Implementation

To use Ball Mapper one only needs three parts: a point cloud, a variable to color by, and an epsilon. For this analysis, our point cloud was made up of 16 variables and 65 points, each representing a year from 1965-2024. Two of the variables, *Unemployment Change* and *Fed Rate Change*, were created before any transformation took place. We elected to create these because the Unemployment Rate (Unemployment) and the Federal Funds Rate (Fed Rate) were the only two dimensions which did not have a transformation and thus did not have a year-over-year change like our other variables. Having the rate of change for both Unemployment and the Fed Rate is also important because it gives us additional information beyond the intrinsic value of each of the measures. Both measures on their own give us a current state at large: the Unemployment Rate telling us how many who are not working but want to,[23] and the Fed Rate signalling the structural cost of capital.[24]

To our knowledge there is no generally accepted way of choosing epsilon in the literature. Since Ball Mapper constructs maps by selecting a point and creating a ball with a radius of epsilon, this is should in theory, given an epsilon small enough, produce a graph such that every point is a node of only itself. Conversely, this should lead to and epsilon big enough such that all of our points are included in one node. For our analysis here, this was indeed the case. So to find an appropriate epsilon we first started by finding lower and upper bounds.

What we look for in our lower bound is a graph with an epsilon small enough make every point in our point cloud a node of one point, a singleton node, not connected to any point. For our upper bound we do the converse, we look for an epsilon so large where it creates a singular node that houses all of our points. Additionally, to reduce some of the computational time one can look for a lower bound where a connected component started to form, and an upper bound where we had a connected component of only two nodes. This cuts off tail end graphs that generally don't provide much substantial insight into the data.

For our analysis we found an initial lower and upper bound represented by the interval $[0.38 - 0.90]$. We then wrote a function to generate around 100 graphs, reviewed them, and narrowed the upper and lower bounds to $[0.40 - 0.70]$. During this narrowing process we were looking for interesting features: connected components forming or dissolving, flares coming off of any components, and notable sizing or coloration patterns. The resulting value of this process was 0.474. We thought it contained all the aforementioned features we look for. This final epsilon is used for all the graphs generated in the analysis.



(a) Ball Mapper graph using standard function, ColorIgraphPlot.

(b) Ball Mapper graph using new graph output created with ggraph.

Figure 1: Ball Mapper graph colored by Year; epsilon set to 0.511.

Before describing some visual changes we made to the Ball Mapper output graph,it is important to note that Ball Mapper will output the same graph unless at least one of two things changes: the size of the point cloud changes or the order of the point cloud. The former intuitively makes sense, the algorithm is creating epsilon radius balls so adding or subtracting any data will yield a different amount of points in at least one ball. Depending on the data being changed or number of points, the structure of the output might not change much. As an example, when we were deciding on which PPI series to use, we had a multitude of choices. Initially we used PPI - All Commodities but found when reading it is susceptible to double counting and used PPI - Finished Goods instead.[18]

The standard Ball Mapper package in R produces a sufficient graph for analysis. The output object of the Ball Mapper algorithm is a series of nodes, edges, edge weights, and other graph properties created by the package *igraph*.[25] Separate from the igraph object, a graphing function that uses R's base plot function and default coloring scheme produces the visual graph. This is an advantage because when packages are written in Base R they are usable to anyone who can install R. We however felt that this graph output could be improved upon and recognized that we could use alternative, and more comprehensive packages to create a graph, *ggraph*.[26]

Using the same underlying output igraph object from Ball Mapper, we created the same graph in ggraph but changed the visual representation and added another helpful piece of information, edge weight, that was given but not used in the default BM graph. Looking at Figure 1, we have the standard Ball Mapper graph output on the left (Figure 1a), and our new ggraph output on the right (Figure 1b). The most prominent changes are the spread of the network and coloration.

Looking at Figure 1b, having the nodes spread out helps us see the structure of our graph more clearly as well as the edge weight between nodes. We see that highlighting the edge strength presents a potential insight between certain nodes. Although our data is relatively small, consisting of 65 rows, one could imagine the advantage Figure 1b might have with a data set much larger that this one (e.g. looking at our time frame on a quarterly or monthly basis). Additionally, we changed the color palette for accessibility and readability.

For Figure 1b and Figure 2, we use Year as our initial coloring variable. We started with this variable because Dłotko, et al.[5] provided a framework for this type of analysis. In this paper the authors started by coloring their graphs by Year to initially understand the shape of their data. Later, in the second half of their analysis, they produced more graphs each colored by the point cloud's variables. Since we have 16 dimensions to our point cloud, an analysis of that size is outside the scope of this paper and is the the subject for another. An additional note about Dłotko, et al., when they colored by the variable 'Year', it was not a variable in their point cloud. Mentioned in the intro, we recognized the usefulness of this technique and extended this idea to our point cloud. Specifically, we calculated variables from out data that we thought represented different economic lenses: the Real Interest Rate (Figure 3), PPI-CPI Gap (Figure 4), and Real Wage Growth (Figure 5).

$$\text{Real Interest Rate} \approx \text{Fed Rate} - \text{Inflation} \tag{4}$$

The Fisher Equation (Equation 4) is a mathematical relationship to approximate the Real Interest Rate. It captures the lived experience of purchasing power for producers and consumers by showing what is left of a nominal interest rate after accounting for inflation–in our case we are using the nominal Federal Funds Rate (Fed Rate). When the real interest rate is negative, inflation is higher than the Fed Rate and is sometimes described as the Federal Reserve being, "behind the curve."[27] Being behind the curve can be from various factors but some common interpretations are, the Federal Reserve does not want to try and tame inflation because of the possible negative impact on the labor market, or the Federal Reserve is hopeful that the economy will grow into the inflation, thereby raising wages up to or beyond inflation. On the other hand, when the real interest rate is positive, the Fed Rate is higher than inflation and could signal that the Federal Reserve is cautious about inflation's impact on the economy. As you might imagine this is described as being "ahead of the curve." If the Fed Rate is ahead of the curve, it indicates the cost of capital is higher causing businesses to "feel" the rate. In reaction to business being more expensive to conduct, employers might cut-cost or freeze hiring, directly affecting the labor market as a result of keeping prices stable. An important note about the Fisher equation, the approximation does not hold well when nominal interest rates (Fed Rate) are relatively high and when time spans are short.[7]

We calculated the PPI-CPI Gap by taking the the difference between PPI - Finished Goods and CPI (inflation). As a quick reminder, PPI represents an "at-cost" revenue (before adding margin, transportation, etc) that producers receive, and CPI is a common measure of consumer inflation. It is important to note that although CPI is used for general consumer inflation, the Federal Reserve prefers Personal Consumption Expenditures (PCE) because it tracks more normal consumer behavior such s substitution of products. The predominate question for this measure was: if producers or consumers are experiencing inflation, can our graph show the primary driving force behind the complex nature that is inflation?

Similar in nature to the the PPI-CPI gap, we took the difference between Compensation and Inflation as an approximate measure for Real Wage Growth. With this measure, we were interested on the relationship between people's annual compensation and inflation,

i.e. are wages were keeping up with inflation. Generally, for this relationship, positive numbers indicate that wages are growing faster than inflation, and vice versa if it is negative–effectively people are taking a pay cut.

RESULTS

Exploratory Data Analysis

Figure 2 shows three different connected components–connected nodes–and many nodes not connected to any others, we will call them singletons. When a node is a singleton, it suggests that the point(s) is unlike all the other data points and indicates that it is a possible outlier in the dataset.[5] Since our graph consists of 40 nodes and three components we will provide all the singletons in a table (Table 4). For the three connected components, we see there is one large and two small components. Moving clockwise, starting with the large component on the left-hand side, we will label them C1,[F] C2,[G] and C3[H] for conversational ease. To prime the reader, the following section will briefly discusses interpretation of Ball Mapper graphs via visual inspection, and then go into further detail about nodes where R is necessary to dissect the contents of each node.

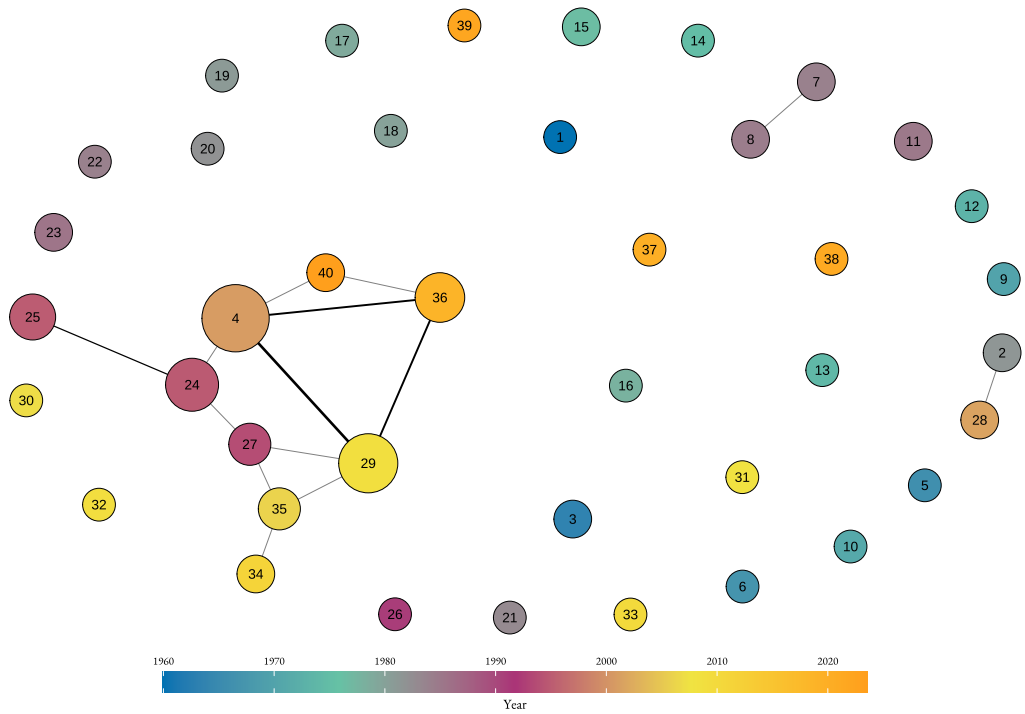


Figure 2: Ball Mapper graph colored by Year; epsilon set to 0.474.

When doing a visual inspection of Ball Mapper graphs, there are essentially two categories to view it through: coloration and shape. Most often coloration of Ball Mapper graph is used to highlight a variable of interest, or certain known structures in the data. This could take the form of highlighting known outliers, show patterning (e.g. a smooth gradient from one end of the graph to the other), or a combination of the two; i.e. nodes which do not conform to patterning. Figure 2, shows us no global coloration pattern, but does reveal four sub-groups of colors in C1 as well as the mismatching coloration for C3. From visual inspection, C1's four sub-groups of colors show distinct time periods: 1990s (magenta), mid-2000s (yellow), 2020s (orange), and 2000 (light terracotta). We also notice that for C3, 2000 and 1980 are what time periods it's nodes show, while C2 shows a consistent mid-1980s coloration. Looking at shape and structure of the graph, as mentioned above, nodes contain an edge between them when there is an intersection of at least one point. The edge weight is commensurate with the number points in two nodes intersection, and the overall size of the node indicates the number

[F] Nodes: 4, 24, 25, 27, 29, 34, 35, 40
[G] Nodes: 7 and 8
[H] Nodes: 2 and 28

of points that nodes has. Particular to C1, nodes which are not directly connected by an edge (e.g. Nodes 24 and 36), can be taken as similar in some fashion, but also signal a significant difference within the component.[5]

Examining C2 and C3 closer, C2's nodes are colored very similarly, and are of the same size. As a relative measure, using Table 4, we look at our singletons and notice that many only hold one point. Thus comparing C2, and C3, nodes we should expect around one to two points in each of the nodes. C2's nodes contain the years 1968 and 1969 in nodes 7 and 8, respectively, and 2000 is in both, creating the edge. Mentioned previously, C3's coloration is not consistent. Node 2 contains 1961 and node 28 contains 2001 with 2002 being our bridge year. What is an interesting observation that will come up again when looking at C1 is how different, but grouped together the years are. This brings up questions as to why 1961 and 2001 would be seen as similar and is explored in the Discussion section.

Structure wise, C1 is the largest component and contains nine total nodes. We mentioned already that it contains three subgroups of colors and observed the indicated years they visually represent. Of the nine nodes the make up C1, we are provided two distinct parts: a large triangle made of the largest nodes with strongly connected nodes (4, 29, and 36); and a sweeping arm to the left of the triangle that contains leaves on each end (nodes 25 and 34).[I] The combination of structure along with the subsets of the colors immediately present a notable result. On the upper half of our arm we see nodes 25, 24, and 27 all colored similar. We have a similar outcome for the bottom half of the arm and one node from our triangle. Conversely, and equally interesting, the triangle's nodes are three distinct colors, while also being the largest nodes in the graph and containing the strongest edge strengths between them.

Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)
1	1960	**10**	1971	**15**	1976, 1977	**20**	1982	**30**	2007	**38**	2021
3	1962, 1965	**11**	1972, 1998	**16**	1978	**21**	1983	**31**	2008	**39**	2022
5	1966	**12**	1973	**17**	1979	**22**	1984	**32**	2009		
6	1967	**13**	1974	**18**	1980	**23**	1985, 1986	**33**	2010		
9	1970	**14**	1975	**19**	1981	**26**	1991	**37**	2020		

Table 4: Singlton nodes and the years they contain.

Although we would need to investigate what points make up the nodes to gain more insight into our data, on surface what we can conclude about C1 is that the sweeping arm seems to have two groups representing two time periods in our range of years. We also notice that nodes 27, 35, and 34 make their own small cycle of nodes but are a mix of the two aforementioned time periods. Lastly, our triangle has a combination of factors which warrant further investigation: the nodes are the largest in size indicating they contain the most points; each pair of the nodes (4, 29, 36) contain the strongest edge strengths of the whole graph; all three nodes are distinct colors indicating separate time periods; and finally, the distinct colors seem to contain a path through time going from 2000 (node 4) around in a counter-clockwise direction to present day (2020-2024).

[I] The term leaf (leaves) could also be called a flare. How we define flare is two or more nodes in a row that are hanging off the main body of a component.

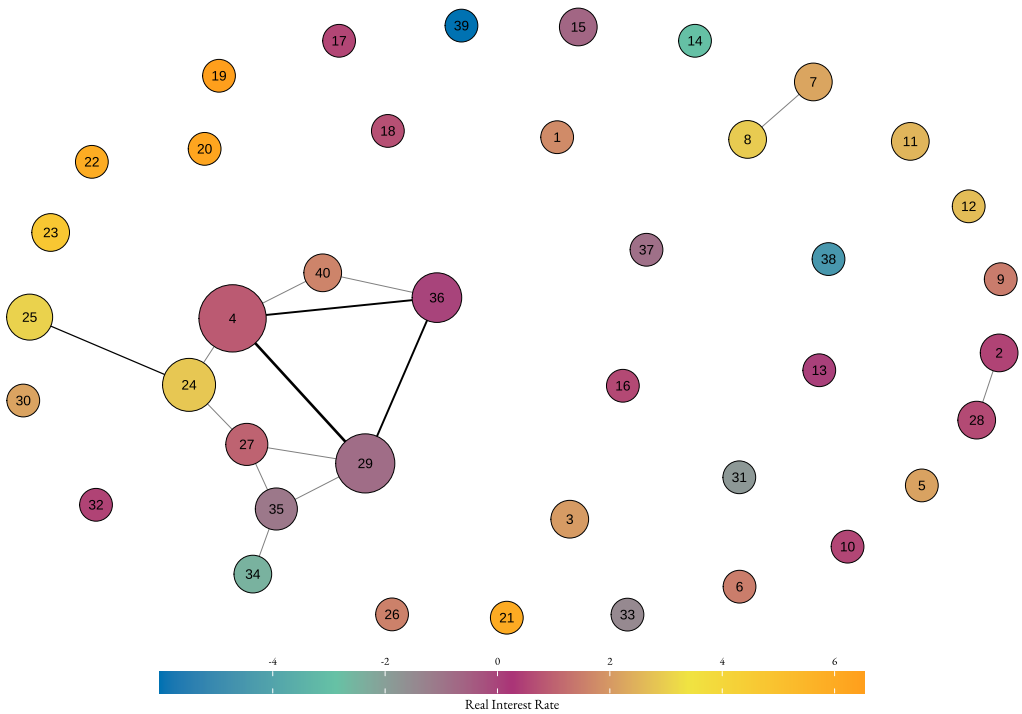


Figure 3: Ball Mapper graph colored by the Real Interest Rate; epsilon set to 0.474.

Real Interest Rate

Figure 3, shows us that colors representing blue/mint green are negative values which represent inflation far outpacing the Fed Rate. In contrast, values colored yellow/orange show us that Fed Rates very high compared to inflation. In general we notice that there are not many nodes which are blue or mint colored, and most nodes range from purple to yellow (approximately -2 to 4). Looking that the extreme ends (less than -4 or greater than 6), we see there are about four nodes for each end. On the low end, nodes 39, 14, 38, and 34; and on the high end there are nodes 19-21. The reader may have noticed both groups of nodes seem to be consecutive (or close to), this is not a random event but a feature in Ball Mapper. We touched on this in the Methods and Procedures: Ball Mapper section, and will revisit this idea in our Discussion. Lastly, and the only case of a coloration gradient, C1 shows a smooth transition from negative values at the bottom (34) to positive in a sweeping fashion like our arm.

PPI-CPI Gap

Figure 4 shows a very different distribution of colors compared to Figure 3. Overall, a majority of our nodes are below the midpoint of the legend. This shows that regardless of the driver behind inflation on both consumers and producers, consumer inflation tends to outpace producers. Additionally, our legend is not balanced like it is in Figure 3; this supports our clustering of values below the middle of our legend. In fact, we see that many nodes seem to be from -2 to -1. With a discerning eye, we estimate to have three nodes which are at the very end of our negative values (nodes 23, 37, and 9). However, on the opposite there is only one node that appears to be an extreme positive, node 39.

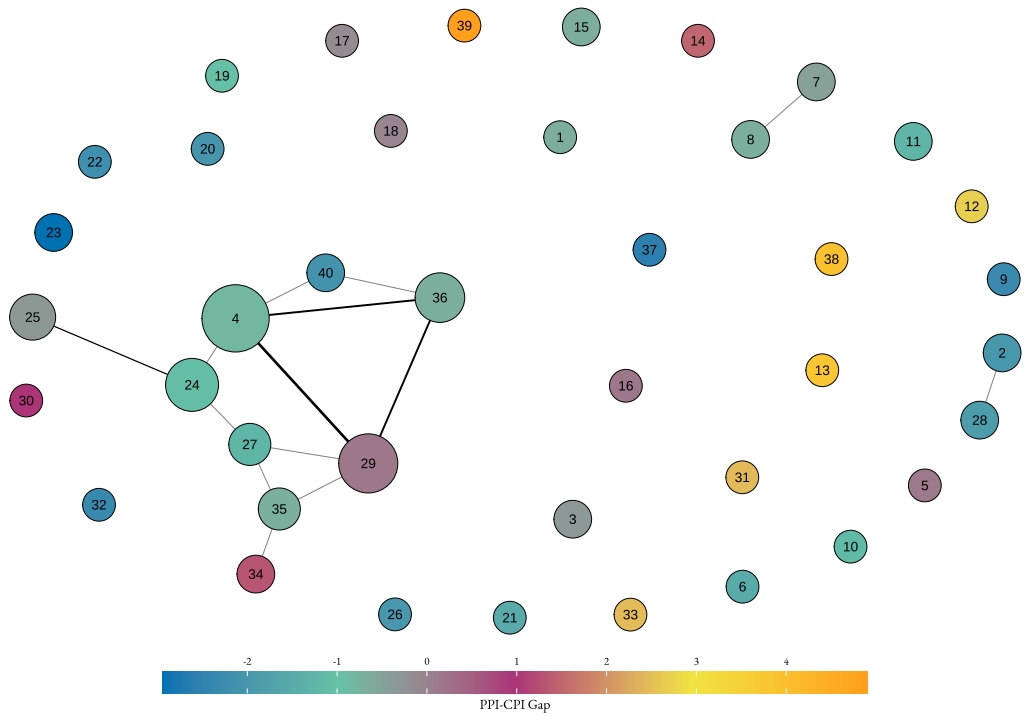


Figure 4: Ball Mapper graph colored by PPI less CPI (PPI-CPI Gap); epsilon set to 0.474.

Wage Growth

Similar to Figure 4, there is a clear concentration of values on one end of our legend Real Wage Growth shown in Figure 5. Using the center of the legend as a reference point, we can see that Figure 5 is even more skewed than Figure 4; its zero starts two tick marks from from the center instead of one. To give a range for the overwhelming yellow coloring, most values seem to fall between -2.5 and 2.5, with most of these values lying right around zero. With many values indicating zero, this means that for our selected time period wages have seemed to keep pace with inflation. Albeit, Figure 5 and Figure 4 are showing different measures, Real Wage Growth shows a much longer tail. Intuitively this makes sense because with the PPI-CPI Gap, both producers and consumers play a similar role, i.e. they are a consumer to someone. On the other hand, if inflation spikes and a business cannot keep up with wages, there can be much more of a divergence. That said, nodes 18, 19, and 17 might be cases for this and further investigation is found in the Discussion section.

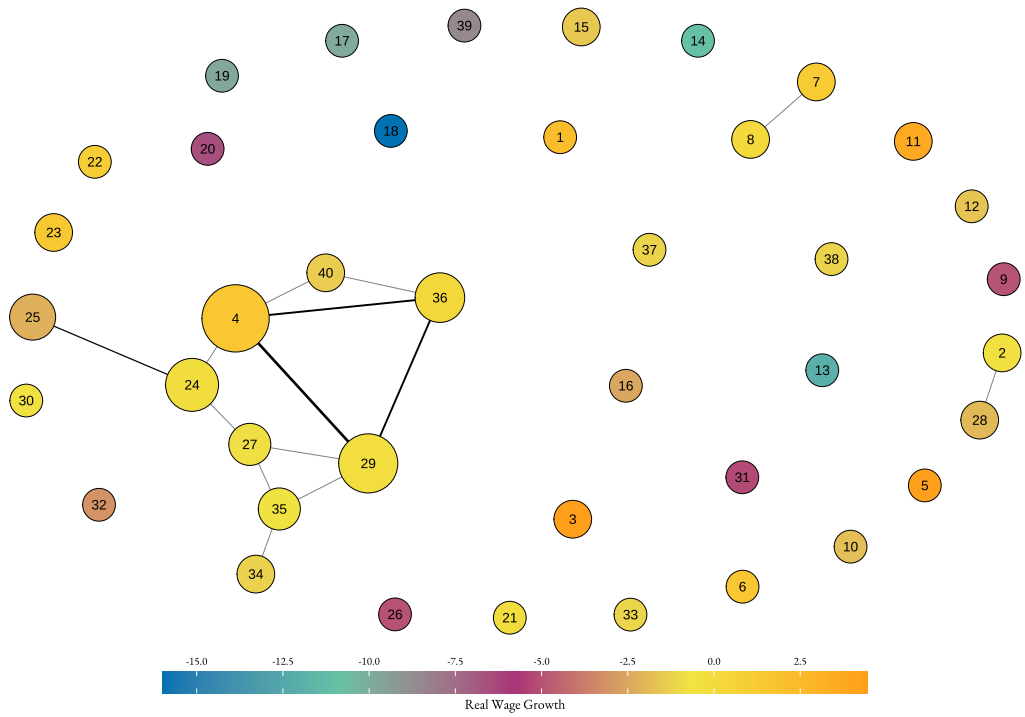


Figure 5: Ball Mapper graph colored by Compensation less Inflation (Real Wage Growth); epsilon set to 0.474.

DISSCUSSION

Often with Ball Mapper analyses EDA and visual inspection are the first steps, and then inspection of what points are housed in each node done after. This process of looking into each node helps us understand why they are seeing certain structures and the coloration. To do this investigative work, we combined all of our dimensions in our point cloud (Table 2) with our three derived economic measures (Real Interest Rate, PPI-CPI Gap, Real Wage Growth), as well as the Year, GDP, NBER recession/expansion indication, and a Row ID to join all the data together (Table 5). For the remainder of the discussion we will look closely at each component, starting with C2 and C3. We will spend a significant portion on C1 due to its size and complexity. Then we will turn to the singletons and finish with some questions and future research.

C2 & C3

C2 is similarly colored across nodes, but this is misleading because of the average value used for coloration. Nodes 7 and 8 contain the years 1968 and 1969, respectively, and the year 2000 forms the connecting edge. Interestingly, Ball Mapper has paired these years together because of their distinct and parallel macroeconomic structures. Our analysis suggests that C2 represents demand-pull inflation, reflecting the adage "too much money chasing too few goods."[28]

As shown in Figure 3 through Figure 5, real interest rates were close to three percent for 1969 and 2000, alongside negative or close to zero values for the PPI-CPI Gap, and positive or near-zero values for Real Wage Growth. These three metrics indicate the economy was running hot but stable. Prices were not rising disproportionately for either producers or consumers. In fact, wages were outpacing inflation, increasing consumer purchasing power.

We also see in our data the late-1960s injection of cash from Lyndon B. Johnson's "Great Society" policies and government spending on the Vietnam War during the mid-1960s.[29] Similarly, in 2000, the private sector was increasing its capital expenditures in anticipation of the digital age.[30] Further, demographic and psychological factors played an additional role in the demand-pull inflation. The 1960s saw the Baby Boomer generation entering their prime working-age years. This expansion in the workforce and subsequent creation of new households increased consumption and lowered unemployment–signs of a healthy economy. The year 2000 presented a similar outcome of high consumption via a different mechanism. Corporate spending was accelerating, which led to high stock valuations,

high market returns, and a general rise in asset values. Consequently, this created a "Wealth Effect", encouraged consumers to increase their spending, placing more upward pressure on demand.[31]

Similar to C2, node 2 is misleading on the surface in Figure 2, it contains years 1961 and 2002 but the average value places it around the 1980s. Node 28 however is much closer in years, 2001 and 2002. Between the two nodes, though, we notice there is a common theme, they represent a structural trough in the economy. What we mean is that this component seems to represent when a contraction has ended, but recovery has not started. Further, unlike C2 we find that C3 shows two different types of troughs: labor market trough and a investment trough. Although both nodes show average muted GDP growth (2.2% for node 2; 1.4% for node 28),

We also observe that in C1, most nodes along with C3 seem to have Real Interest Rates showing between -1% and 2%. To contrast this, we see that nodes 25, 24, and C2 look to be above 2% but not above 4-5%.

end c3 section

As mentioned above, what is striking about C1 is that there are two shapes: a triangle and a sweeping arm that runs along the triangle's left side. Although we have not found any literature that explores the relationship between nodes and edge strength (how many points they share), the triangle (denoted body from now on) was something immediately of interest. Beginning with Real Wage Growth, it falls between -2.5 and 2.5 (Figure 5), with an average value of -0.065. A value close to zero tells us that for the points in C1, wages have generally kept up with inflation. The PPI-CPI Gap (Figure 4) shows there is a main group of nodes centered around negative one with three nodes higher (25, 34, and 29), and one lower (40). These mostly negative values suggest that consumer inflation is generally outpacing producers in C1. It's worth noting, however, that our PPI measure does not consider profit margin added, transportation cost included, or other on-the shelf costs. So this small differential that we see in C1 possibly could be reflecting that gap in pricing, but a more in depth analysis or additional data in our point cloud is needed to provide a more definitive answer. The real interest rate graph, Figure 3, displays a color gradient for C1 that immediately stood out to us. Seeing that it moves from negative to positive values in a semi-orderly manner supports the idea that the nodes with negative values show inflation was outpacing the nominal Fed Rate; and as we move toward the positive values, we see that relationship flip, with the Fed Rate exceeding inflation. Supposing that the Federal Reserve was behind the curve for the negative values and ahead for positive, along with our knowledge that overall real wages have kept up and the PPI-CPI Gap is relatively small, we start to get a picture of what C1 could represent.

Indeed, what we have found for C1 is that it represents the U.S. economy at different states of growth. This does not mean that every node in C1 is an instance of good economic conditions, but rather nodes represent certain "phases" that the economy goes through until it turns into a state of overheating expansion (peak) or coming out of a state of contraction (trough). What we have also found is that these peaks and troughs are not included in C1, but are represented by C2 and C3 respectively.

Broadly, the stages can be categorized in different phases: recovery (34 and 35), stability (29, 4, and 36), resistance (24, 25), and non-canonical (27 and 40). Starting with our recovery phase, we find signs of an economy that has been beaten down but is starting to come back to life. We find that 34 shows a very fragile economy and 35, being connected to more nodes, indicates more stability. The first thing we notice is that housing starts is positive, so builders are feeling confident that they can start projects. The "big four" economic indicators–GDP, Inflation, Unemployment, Fed Rate–on average, show signs of improvement. GDP is increasing, inflation cooling, Unemployment starting to fall, and the Fed Rate relatively low because they are trying to stimulate the economy. Additionally, real interest rates are negative and real wage growth is improving, suggesting that producers and consumers feel more confident about the economy.

Next, we have what we call the resistance phase. Unlike our recovery phase, nodes 24 and 25 show two different sides of resistance. Node 24 shows an economy which feels like it is still growing and running well. Unemployment, on average, is 5.4% and trending downwards, inflation seems to be under control, around 3.1%. However, at this stage there seems to be some cracks forming in the economy, the first being housing starts turning negative. What is also worrisome is the high Fed Rate (5.9%) and the negative PPI-CPI gap (-1.1%). The combination of these suggests that the Federal Reserve sees that CPI is running ahead of PPI and are concerned about producer inflation catching up to consumer inflation. With CPI outpacing PPI, consumer prices are currently rising, however if producer costs start to rise, the PPI-CPI Gap will start to close and inflation then is a problem for both producers and consumers. To counteract this, the Federal Reserve is preemptively trying to kill inflation for consumers with higher rates. Unfortunately, this has knock-on effects to the labor market restricting wage growth and possibly reversing the direction of unemployment. Node 25 shows the other side of what 24 was warning us about. Housing slows down even more (from close to -5% to -10%, on average), and the PPI-CPI Gap shrinks to almost zero; unlike node 24, producers are now starting to feel cost-push inflation. Since both producers and consumer prices are rising, the Federal Reserve continues to raise rates to an average 7%, sending average real interest rates above 3% and risking the labor market to decline and the economy to possibly enter a recession. We also see real wage growth plummeting from -0.16% (node 24) to -2.23% (node 25). To illustrate this differential, suppose we were given $1,000 at the end of the year and to either had to give some back (negative wage

growth), or we received extra (positive wage growth). In node 24 we get that $1,000 and give back $1.60, however in 25 we would have to give back $22.30. So, node 25 shows a "last resort" node before turning into a singleton node. This is similar to 34 in the sense that it is only connected to the other nodes in C1 by one other node, supporting this idea that they are the exit and entrance into economic growth.

Year	PPI Finished Change	Personal Income	Compensation
Entrepreneurship	Transfers	GDP	Consumption
Domestic Investment	Government Spending	Exports	Imports
Inflation	Unemployment	Unemployment Change	Fed Rate
Fed Rate Change	Housing Change	Recession Span (0-1 by 0.25)	Expansion Span (0-1 by 0.25)
Row Number	Real Interest Rate	PPI CPI Gap	Real Wage Growth

Table 5: Data used to review nodes, included additional data that was excluded from out point cloud.

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INCOMPLETE WRITING STARTING HERE

Notes for C1

Node 4 shows us an "ideal" version of the economy. We have personal income and compensation slightly higher than inflation, sitting around 2%. Unemployment is a little elevated but historically not unusually high.

Node 24 shows us years where the economy is starting to get "over it's skis". That is, it is starting growing faster than it can keep up with but it's not necessarily too late.

Node 25 on the other hand consists of years following right after recessions or points where the economy was over heating. Most of the years in Node 26, follow what economist call a "Jobless Recovery".[32] Usually marked by growth returning to the economy but unemployment being "sticky" and not falling as business starts to take off.

Singletons (contraction side of business cycle)

Singleton nodes 19-22 have relatively high values (average of 6.178%) consist of years 1981-1984. This directly lines up with the experience "Volcker Shock".[33]

Singleton nodes 19-22 have relatively high values (average of 6.178%) consist of years 1981-1984. This directly lines up with the experience "Volcker Shock".[33] On the opposite end, moving from the twelve o'clock position clockwise, we can visually see nodes 39, 14, 38, 31, 33, 34 (2022, 1975, 2021, 2008, 2010, 2011-2012 respectively) show a negative value (Equation 4).

When we look at the coloration of the singleton nodes, we see indication of outlier events such as The Great Recession and the COVID-19 Pandemic.

Further this short time period not only is represented by interesting about why these nodes are colored is that these nodes see to show the impact of Volker's decision to raise interest and the lag that took place when raising rates.[34]

In these year rates were already extremely low, or the economy just came off a huge economic shock. In 1975 the fed aggressively lowered rates due to climbing inflation and high unemployment rates from the 1973-'74 oil embargo.[35] In 2021-'22 the COVID-19 Pandemic stimulus checks were going to cause inflation but the fed did not want to raise rates too aggressively.

Positive values indicated that businesses are having their revenue rise higher than consumer inflation. This could mean various things: corporate margins are expanding, they are passing through cost to customers, or....

39, 39, and 13 (2022, 2021, 1974) show high positive values indicating higher PPI values relative to Inflation. On the opposite end, the nodes which represent highest inflation relative to PPI values are nodes 23, 22, 20, 37, 9, 2, 28, 26, and 32 (1985-86, 1984, 1982, 2020, 1970, 1961 and 2002, 2001-02, 1991, and 2009 respectively).

Looking at nodes 19, 17, 18, 39, 14, and 13,[J] we see that these are the worst in terms of inflation outpacing people's compensation. Indeed, when we look at the years contained in the nodes, they are years which had record breaking inflation - The 1970s Oil Embargo, The Volcker Shock, and 2022 once all the stimulus cash made its way through the economy. On our positive end we have nodes 11, 5 and 3 which consist of years 1972 and 1998, 1966, as well as 1962 and 1965. What is interesting that we don't see in Figure 3 or Figure 4 is that the legend shows a long tail with our zero point being on the far right end. Notice, most of our points are shades of yellow (around zero or little below) but we have very few values that represent negative double digits.

Future Areas of Work/ Consdierations

Talk about the semi-randomization of BM. If any data in the point cloud is modified in anyway, this ball creation process will change..

Robustness in data

What is unclear, and unstudied to our knowledge, is if the path length between two nodes is a function of similarity.

uknown if distances on our graph indicates something? 34-35 path is small and logical recovery. 24-25 is large but the resistence is quite dis jointed.

This also applies the nodes of large size that are connected. We have not found any literature that supports this idea that proximity is commensurate with relatedness.

Analysis showed only 65 years of data and nodes contains some points which were structureal at the time. smaller time periods might show more robusntes and that the Year or time period is less of a factor than what we might ifnd here.

Look at sitribution of nodes to each, if spacing is meaningful or not

Relationship of years that creat edges signal that throughout that year the economy went through various stages

For the future development of TDABM, and TDA in general, there are many possibilities of future research and one glaring downside. The downside to TDA is that it is a high barrier to entry to understand the methodologies inner workings. Those with a technical background will have an easier time, but nonetheless a higher barrier than methods such as linear regressions. Subject wise, TDA has an advantage in some of the social sciences because it can be somewhat of a bridge between qualitative and quantitative analyses. TDABM has been used to analyze the Brexit Vote data, seeking to understand quantitative and qualitative motivations behind its outcome. TDA also has the advantages of dimension reduction which could be useful to the life-sciences. Fields such as genetics, chemistry, and biology have many dimensions of data, and being able to consolidate it into an understandable form could benefit the body of knowledge greatly.

Unused, so far

Another way to view this is, clustering revels that the financial crisis was a singular problem that spurred the collapse of the financial system. Conversely, the COVID pandemic was a singular issue but exposed many different weaknesses in our current world.

CONCLUSION

Although there is much further work to do on the developing Ball Mapper, through our analysis we have show how TDA, and Ball Mapper in particular, is able to give insight into data that we might not see or describe easily. Throughout our analysis we have hopefully guided our reader to better understand how Ball Mapper works and how to utilize it. We also have show a structural map of the U.S. economy from 1960 to 2024.

C1 structural expansion. C2 structureal "clean" peak. C3 controlled and clean trough. singltons corrections/contractions, all uniquely drivien.

As seen in many other, longer papers, using app written by Dlotko and colleagues, there is a lot of room to expound our TDABM graph here,though that is beyond the scope of this paper.

In general, though,

Consider our comparison of the 08' - 09' financial crisis and the 2020 COVID Pandemic: We see learned that the years following the financial crisis were similar economically.

[J] 1981, 1979, 1980, 2022, 1975, 1974 respectively

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ABOUT THE STUDENT AUTHOR

Ryan Z. Johnson is an alumnus of the University of Alaska - Anchorage. They graduated spring of 2018 with a Bachelor's of Science in Mathematics. During their senior year, Ryan was urged by their advisor Dr. Mark Fitch to pursue TDA after doing presentation for a course and connected with Dr. Samuel Cook. Since then, they have continually worked with Dr. Cook outside their profession in social services. They will be continuing their education at the University at Albany as a graduate student in the Master's in Data Science

program. This paper provided valuable preparation and experience for the upcoming program's concentration areas: TDA, machine learning, and statistics.